IT159: Artificial Intelligence
Lab#1/Assignment#1: Designing Vacuum Agents

1. Introduction

The "agent" paradigm has rapidly become one of the major conceptual frameworks for understanding and structuring research activities within Artificial Intelligence. In addition, many software frameworks are being introduced and used both in research and industry to control the complexity of sophisticated distributed software systems, where nodes exhibit behaviours often associated with intelligence.

The purpose of this lab is to introduce the agent concept by allowing you to familiarise yourself with a software agent environment and world simulator. You will program an agent and implement its behaviour to provide cleaning capabilities in the vacuum cleaning world. The behaviour will then be compared to some simple reference agents.

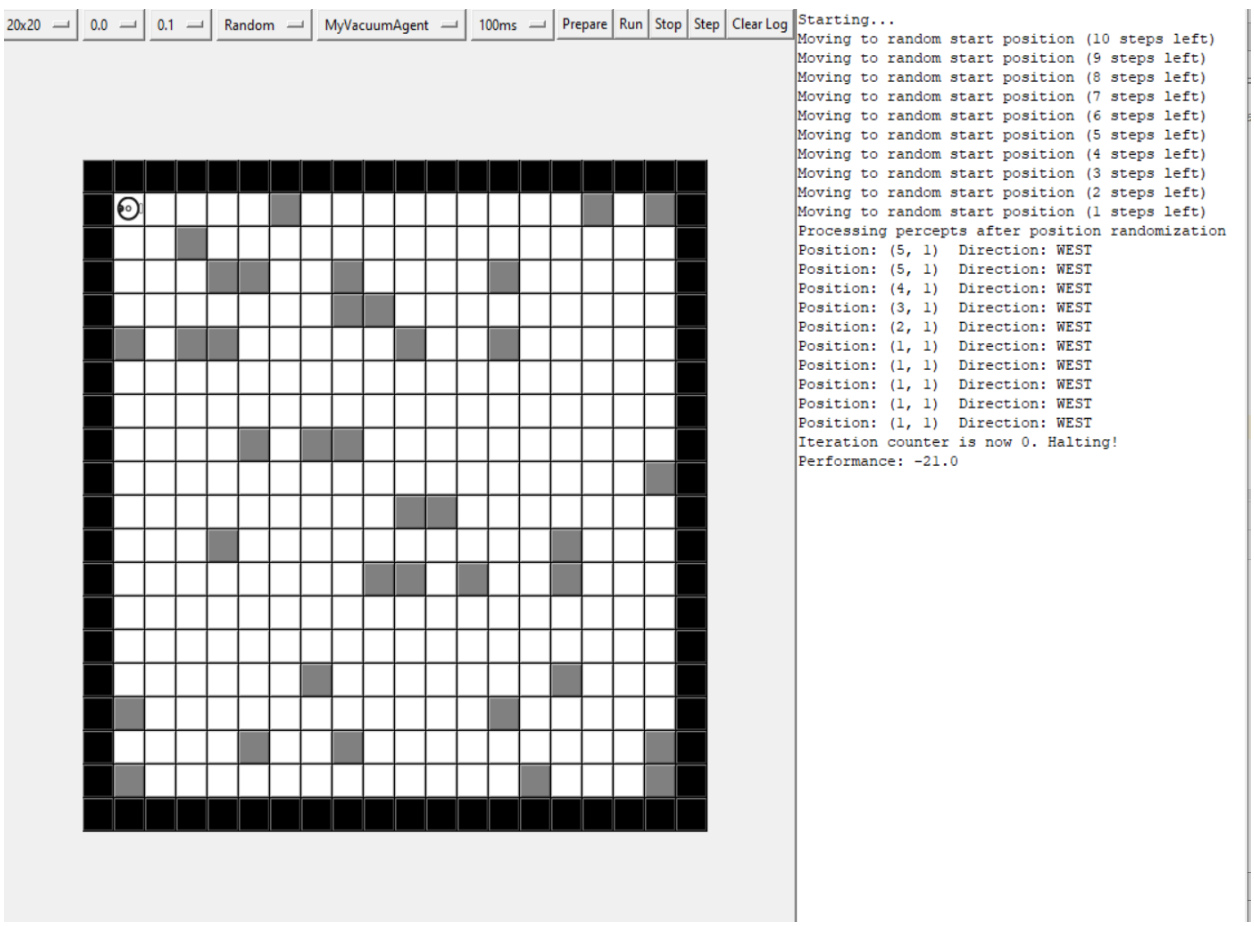

Figure 1: *Vacuum agents' world.*

2. Lab assignment

Your assignment is to construct a vacuum agent. The agent should be able to remove all the dirt in a rectangular world of varying size with a random distribution of dirt. When the world is cleaned, the agent should return to the start position and shut down.

3. Lab system

The lab system consists of the environment simulator and the agent. You are given a simple example of a vacuum cleaner agent called `myvacuumagent.py`. What you have to do is improve the code controlling the agent's behaviour to make it perform better. The instructions for setting up and running the code are available on Blackboard.

The environment simulator

You can test your agent in *the simulator GUI* program. The simulator loads an agent (yours will be implemented in `myvacuumagent.py`) and executes it in a generated environment. Agents execute actions in the environment depending on what they can observe (perceive). The actions influence the environment, which in turn generates new sensory impressions for the agents. The simulator handles this interaction in a discrete manner; the world advances in steps, where each step can be described as follows:

1. The agent perceives the world
2. The agent decides on an action
3. The action is executed

The agent should be autonomous; therefore, the simulator makes no assumptions about the agent except that it can produce a reaction based on the sensory inputs presented to it. The agents make no assumptions about the simulator's implementation either. This disengagement is important, as it enables comparison and evaluation of different agents' behaviours. Note that an agent has no authorisation to change the world directly, for example, via some global reassignment. This disengagement keeps the code modular and independent, allowing the representation of the world to change while keeping the same agent.

The implementation of the simulator

The simulator supplied for this lab is based on the AIMA-Java/AIMA-Python framework, but it is not identical to the original version available on the web.

The interface

Basically, the agents may perceive dirt or obstacles in the squares, and they can act by moving between the squares and removing dirt.

Percepts

A percept for the vacuum-agent consists of three values:

1. bump equal to true if the agent has bumped into an obstacle, else false.
2. dirt equal to true if the current square contains dirt, else false.
3. home equal to true if the agent is at the start position, else false.

The three Boolean values are obtained with the following code located in the execute method of an agent:

```
bump = percept.attributes["bump"]
dirt = percept.attributes["dirt"]
home = percept.attributes["home"]
```

Note that this is the only data that the agent gets from the simulator.

Output

The output from the agent's program (return value) is one of the following actions:

1. ACTION_FORWARD moves the agent one step forward depending on its direction,
2. ACTION_TURN_LEFT makes the agent turn left.
3. ACTION_TURN_RIGHT makes the agent turn right.
4. ACTION_SUCK cleans the dirt from the agent's current location.
5. ACTION_NOP informs the simulator that the agent has finished cleaning.

GUI

The provided environment simulator application is presented in Figure 2.

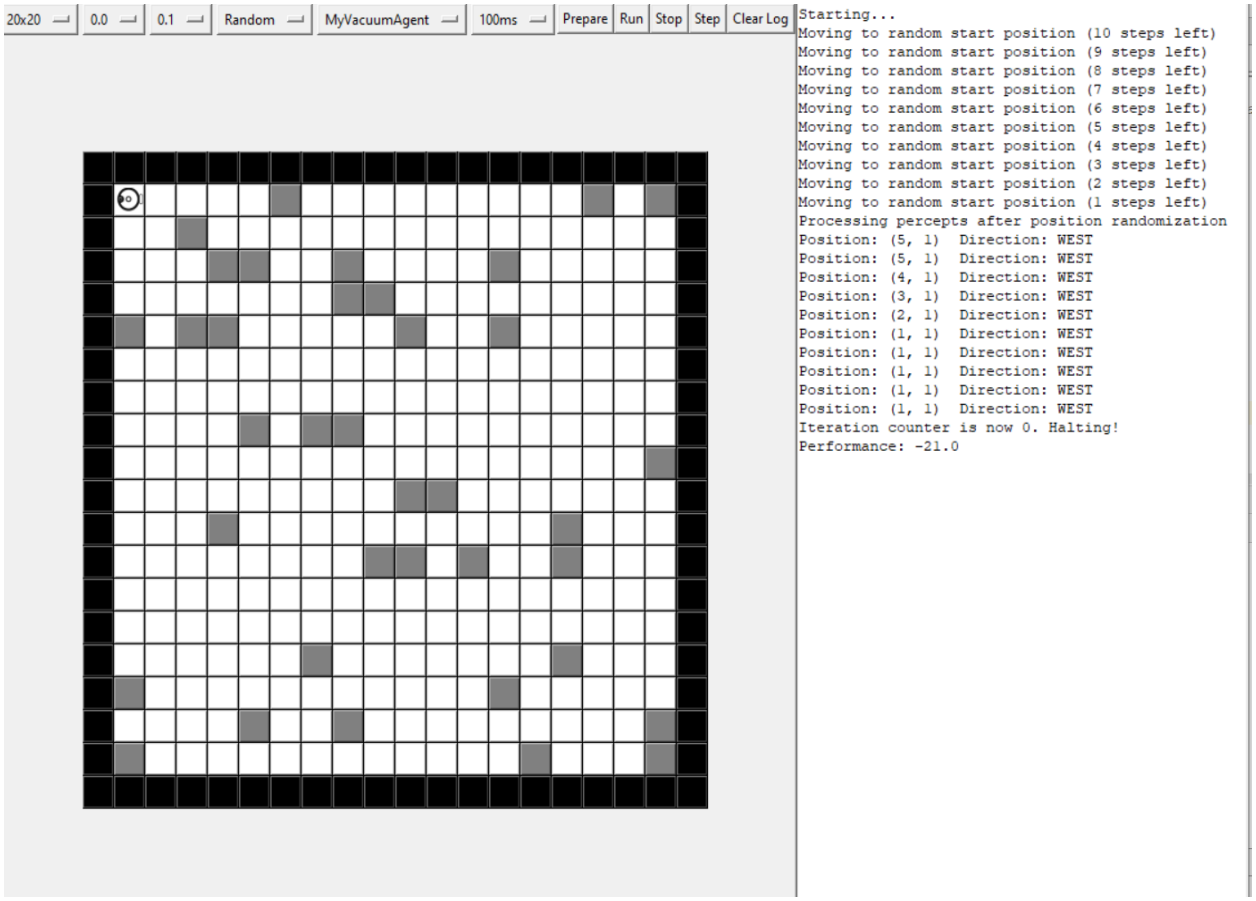

Figure 2: *Example of an environment in the vacuum agent's world*

The main window of the application is divided into two parts: the environment visualisation pane on the left and the text output pane, which displays the redirected *self.log* in Python. Certain messages from the simulator are printed in this pane, but it is also useful to debug your agent's code.

In the environment pane, the black squares represent obstacles. Trying to move an agent into an obstacle results in perceiving the bump equal to true. The grey squares represent the dirt. Moving into such a square results in perceiving the dirt equal to true. The graphical representation of the agent visualises its current direction.

The following drop-down menus are available to set the simulation parameters (from left):

1. *Environment size* can be set to *5x5, 10x10, 15x15, 20x20, 5x10, and 10x5*. Your agent should not be tailored to any specific size, though for convenience, you can assume a maximum size of *20x20*.
2. *Hinder probability* indirectly specifies the number of obstacles in the environment (it should be set to 0.0 and 0.1, in Tasks 1 and 2, **respectively**).
3. *Dirt probability* indirectly specifies the amount of dirt in the environment.

4. *Random environment* specifies whether a new environment should be generated for every run. This function is useful for testing an agent in the same environment multiple times.
5. *Type of agent* lists the available agent type classes. The first on the list is the agent you will modify.
6. *Delay* specifies the delay between each simulation step.

The following buttons are available (from left):

1. *Clear* cleans the text output pane.
2. *Prepare* generates a random environment based on the selected parameters.
3. *Run* starts the simulation.
4. *Step* allows for performing one step of the simulation at a time. Make sure to execute Prepare first.
5. *Cancel* terminates the simulation.

Program

The interesting part of the agent is the execute method. It is a function that takes a percept as input and returns an action as output. The simulator computes what the agent can perceive (a percept), applies the function (your program) to the percept, and executes the resulting action. Note (again) that the agent has no authorisation to change the world directly, for example, via some global reassignment.

As you can see from the definitions in the source code, the *RandomVacuumAgent* and *ReactiveVacuumAgent*, do not remember what they have done and experienced. If we want the agent to have a memory, we have to store information "inside" the agent.

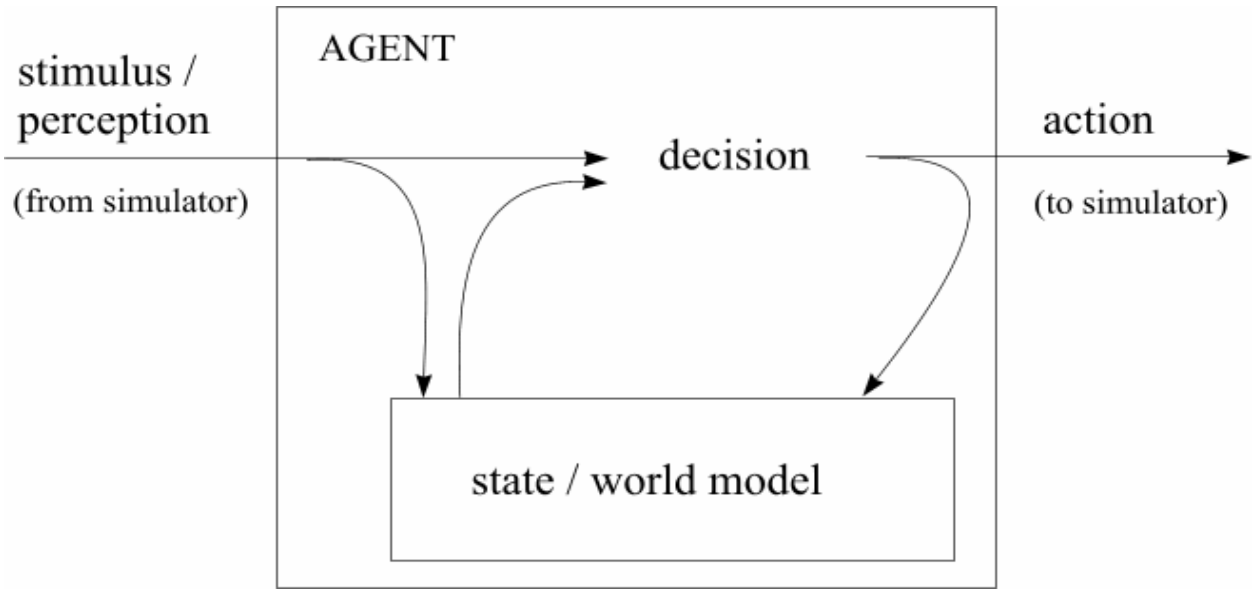

Figure 3: Agent with internal state

The state can be a model of the surrounding world as well as some other knowledge that the agent has, e.g. plans.

Scoring

The performance of an agent is measured by a score. The agent's initial score is -1000, and the score changes with each action. Sucking up dirt increases the score by 100 points, while shutting down at the starting square gives it 1000 points. All other actions decrease the score by one point.

4. Running the Lab environment

You can use your editor of choice (e.g. PyCharm) and follow the instructions below. Note, the lab code requires Python version lower than 3.10 and depends on three additional modules: numpy, tkinter and ipython.

1. Console: to run the code execute the following command in the Lab1 directory:

   python3 run_lab1.py

2. PyCharm: setup and start the PyCharm editor:

   - To add the PyCharm editor to your environment execute the following command in the terminal: module add prog/pycharm-professional/2021.2.3 or alternatively module initadd prog/pycharm-professional/2021.2.3 to always add it when logging in - note that you will need to open a new terminal window for the command to take effect the first time.
   - pycharm.sh
   - After starting the PyCharm editor, open the project by selecting File/Open and choose the Lab1 directory.
   - To execute run run_lab1.py file - open the file in the editor, right click and choose Run run_lab1.

5. Detailed Lab Task Description

Play around with the GUI and try the different options and agents.

1. Using myvacuumagent.py as a template, implement an agent with an internal state that helps the agent clean more effectively/efficiently than the reactive/random agents. Note that an example of an agent with a simple state representation is provided (very limited - it only moves forward and sucks dirt) that you may modify in any way you wish. The agent must be able to suck up all dirt in a rectangular world of unknown dimensions**, without obstacles, and shut down at the home position (**the home position can be sensed through one of the percepts). At the end, the agent should also have a fully updated world model (i.e. world variable in self.world).

Note: The agent starts from a random location, which is achieved by executing the method in the move_to_random_start_position in Python. You should not change this part of the code. In Python, simply edit myvacuumagent.py file and run the lab again. Evaluate the performance of the agent. In what ways is it better or worse than the reactive agent?

2. In this task, you will extend and improve your agent to be able to solve the problem **with obstacles** (0.1 obstacle density, 15x15 world). This can be achieved in a variety of ways with different levels of complexity (e.g. rule-based, search). Returning to the **home** position is not strictly required, but encouraged. It is sufficient that the agent performs a predefined minimum number of steps (set **self.iteration_counter** - Python variable to e.g. [width x height x 2]) before shutting down. Prepare to discuss during the lab demonstration how you have improved the agent's ability to handle obstacles.

   Note: If you are already familiar with search algorithms, feel free to implement an agent based on search which shuts down at the home position after all the dirt is cleaned. In that case, the iterationCounter should be set to a sufficiently large value so that the agent can finish the task.

3. Write explanations of how your agent works (in both Task 1 and 2), what is the main decision "loop", some notes about what data you store in your agent, and why you have chosen your solutions, if there were alternatives, etc. This should add up to a report of approx. 1-2 A4s. Commented or easily readable code is also required.

What to submit:

Your submission should include the following:

1. Lab report answers from Task 3 and source code + README (how to compile and run your code).
2. Please create a folder called "yourname_studentID_Lab1" that includes all the required files and generate a zip file called "yourname_studentID_Lab1.zip".
3. Please submit your work (.zip) to Blackboard.